CIE AS PHYSICS (9702) • MARK SCHEME & MODEL ANSWERS

7.2 Transverse & Longitudinal Waves

Syllabus Section: 7. Waves **Paper Type:** Structured Questions (Paper 2) **Total:** 5 Questions (44 Marks)

Question 1 Easy [7 marks]

Part (a) [1 mark]

Define a transverse wave.

Mark Scheme (1 marks):

A wave is transverse if...

- The oscillations are perpendicular / at right angles to the direction of wave motion / energy transfer; *[1 mark]*

[Total: 1 mark]

Part (b) [2 marks]

State **two** properties of a longitudinal wave.

Mark Scheme (2 marks):

The properties of a longitudinal wave are:

*Any **two** from:*

- The oscillations are parallel to the direction of wave motion / energy transfer; *[1 mark]*
- Comprises of areas of compressions and rarefactions; *[1 mark]*
- Requires a medium to propagate; *[1 mark]*
- Cannot be polarised; *[1 mark]*

[Total: 2 marks]

Part (c) [2 marks]

State **two** examples of transverse waves.

Mark Scheme (2 marks):

Examples of transverse waves are:

*Any **two** from:*

- Water waves; *[1 mark]*
- Secondary waves / S-waves; *[1 mark]*
- Vibrations on a (guitar) string; *[1 mark]*
- Electromagnetic waves / any named EM wave (gamma, X-ray, ultraviolet, visible light, infrared, microwave, radio); *[1 mark]*

[Total: 2 marks]

Part (d) **[2 marks]**

State **two** examples of longitudinal waves.

Mark Scheme (2 marks):

Examples of longitudinal waves are:

*Any **two** from:*

- Sound waves; *[1 mark]*
- Ultrasound; *[1 mark]*
- Primary waves / P-waves; *[1 mark]*

[Total: 2 marks]

Question 2 Easy [6 marks]

Part (a) [2 marks]

Fig. 1.1 shows graphs of waves X and Y. The scales of the graphs are identical.

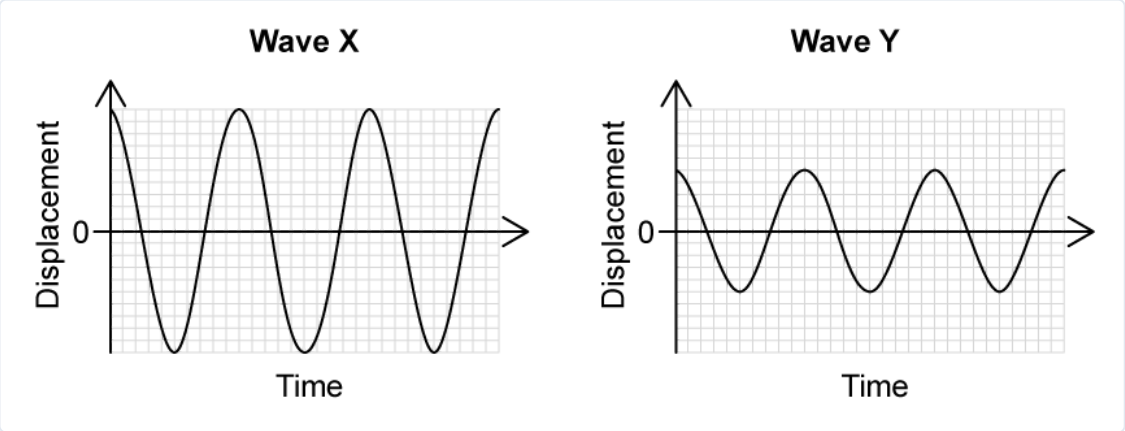


Fig. 1.1

Describe **two** similarities between the properties of waves X and Y.

Mark Scheme (2 marks):

The similarities between waves X and Y are:

- Both waves have the same period (10 squares); *[1 mark]*
- Both waves have the same frequency (3 complete waves in the time shown); *[1 mark]*

[Total: 2 marks]

Wavelength would not be an acceptable answer here, because the graph shows time on the x-axis not distance, therefore the distance between each trough or crest is the period.

Part (b) [1 mark]

State **one** difference between waves X and Y.

Mark Scheme (1 marks):

The difference between waves X and Y is:

- The waves have different amplitudes **OR** wave X has a greater amplitude than wave Y; *[1 mark]*

[Total: 1 mark]

Part (c) [2 marks]

Distinguish between transverse and longitudinal waves.

Mark Scheme (2 marks):

Transverse waves are:

- Vibrations / oscillations that are perpendicular / normal to the direction of energy transfer; *[1 mark]*

Longitudinal waves are:

- Vibrations / oscillations that are parallel to the direction of energy transfer; *[1 mark]*

[Total: 2 marks]

Part (d) **[1 mark]**

Explain how it could be determined whether waves X and Y are transverse or longitudinal.

Mark Scheme (1 marks):

The wave type can be determined by:

EITHER

- Determining the motion of the particles relative to the direction of propagation / energy transfer; *[1 mark]*

OR

- Polarisation since only transverse waves can be polarised; *[1 mark]*

[Total: 1 mark]

Question 3 Medium [10 marks]

Part (a) [2 marks]

Explain what is meant by a progressive wave.

Mark Scheme (2 marks):

Explain what is meant by a progressive wave:

- A transfer of energy; *[1 mark]*
- With no transfer of matter; *[1 mark]*

[Total: 2 marks]

Part (b) [4 marks]

An elastic cord is fixed at one end and attached to a mechanical oscillator at the other end. Fig. 1.1 shows, at time t = 0, the shape of a section of the cord as the wave travels from left to right. W, X, Y and Z are four marked points on the cord.

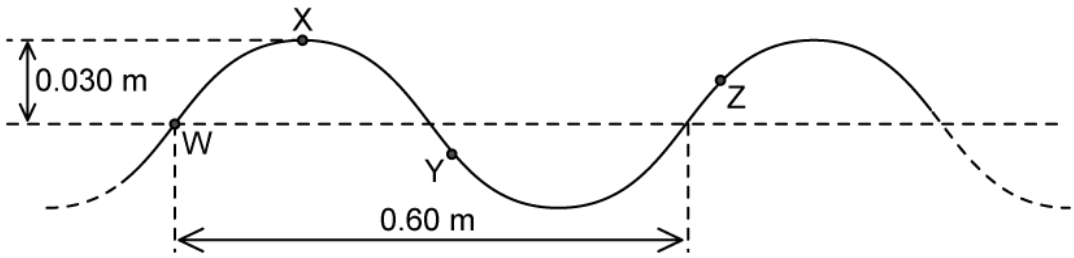


Fig. 1.1

The mechanical oscillator has a steady frequency of 5.0 Hz. The wave has a wavelength of 0.60 m and an amplitude of 0.030 m.

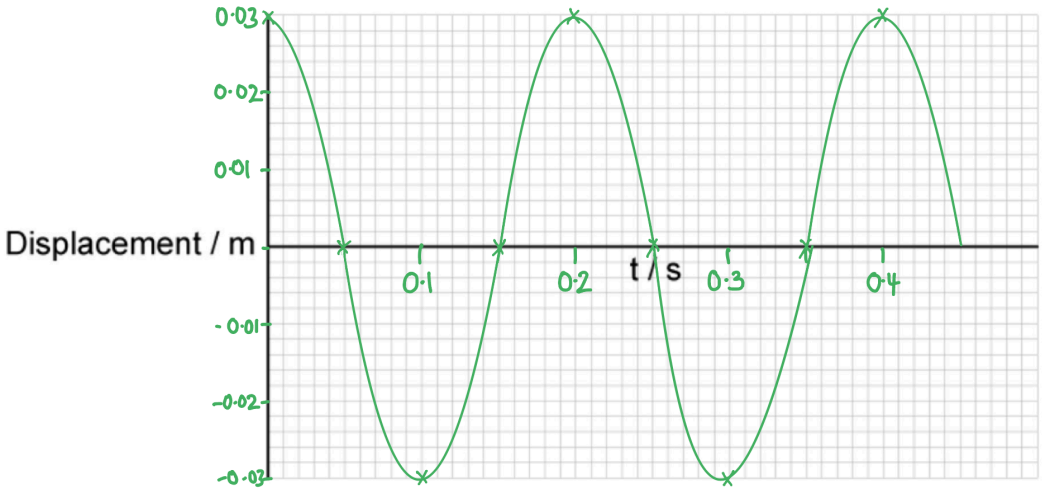
On the axes of Fig. 1.2 sketch the graph of the displacement of point X over the period t = 0 to 0.40 s. Add suitable scales to the axes.



Fig. 1.2

Mark Scheme (4 marks):

A correctly completed graph of the oscillation looks like:



- Shape is sinusoidal with exactly two wavelengths shown; *[1 mark]*
- Amplitude at ± 0.03; *[1 mark]*
- Period = 0.2 s **AND** x-axis labelled 0.1, 0.2, 0.3 and 0.4; *[1 mark]*
- Curve starts at (0, 0.03); *[1 mark]*

[Total: 4 marks]

Part (c) **[2 marks]**

For time t = 0, state which of the points W, X, Y and Z

i) is instantaneously at rest

[1]

ii) has the greatest speed

[1]

Mark Scheme (2 marks):

For time t = 0, state which of the points W, X, Y and Z:

i) Is instantaneously at rest:

- X; *[1 mark]*

ii) Has the greatest speed:

- W; *[1 mark]*

[Total: 2 marks]

Part (d) **[2 marks]**

The speed of point W on the cord at t = 0 is 0.94 m s^{-1}. With the cord at its original tension, the frequency of oscillation is now doubled to 10 Hz.

The amplitude is kept at 0.030 m. Calculate the new speed of point W at t = 0. Explain your reasoning.

Mark Scheme (2 marks):

Calculate the new speed of point W at t = 0. Explain your reasoning:

- Frequency has doubled so point W has got to move double the distance in the same time **OR** move the same distance in half the time; *[1 mark]*
- Therefore speed doubles OR new speed = 1.9 m s^{-1} *[1 mark]*

[Total: 2 marks]

Question 4 Medium [12 marks]

Part (a) [3 marks]

Explain briefly how transmission of energy by a longitudinal wave differs from transmission of energy by a transverse wave. Give **one** example of a transverse wave.

Mark Scheme (3 marks):

Explain:

- The disturbances/oscillations/vibrations are parallel for longitudinal and perpendicular for transverse *[1 mark]*
- To the direction of transmission of energy/wave/propagation/direction *[1 mark]*

Do not accept 'movement' of the wave

Example:

*Any **one** from:*

- Electromagnetic waves e.g. microwaves, radio waves, UV *[1 mark]*
- Waves on a string *[1 mark]*
- Seismic S-waves *[1 mark]*
- Ripples on the surface of water *[1 mark]*

[Total: 3 marks]

Part (b) [3 marks]

With the aid of a clearly labelled diagram explain how a sound wave in air transmits energy away from its source.

Mark Scheme (3 marks):



- Clear diagram of particle oscillations / pressure variations (compressions and rarefactions)

AND

- Includes label indicating direction of propagation; *[1 mark]*

Explain:

*Any **two** from:*

- The sound source is a vibrating source (e.g. speaker); *[1 mark]*
- Energy is transferred to (air) molecules; *[1 mark]*
- Energy is transferred by collisions between molecules; *[1 mark]*
- Oscillations of (air) molecule neighbours are slightly out of phase; *[1 mark]*
- Oscillations / waves are longitudinal / energy transfer is parallel to vibrations; *[1 mark]*

[Total: 3 marks]

Part (c) [3 marks]

Short pulses of sound are reflected from a wall 30 m from the sound source. The reflected pulses return to the source after 0.18 s.

Calculate the speed of sound.

Mark Scheme (3 marks):

The total distance is equal to the distance to and from the wall

- Distance travelled = 2 × 30 = 60 m *[1 mark]*

Calculating the speed of sound:

- Speed = $\frac{\text{distance}}{\text{time}}$
- Speed = $\frac{60}{0.18}$ *[1 mark]*
- Speed = 333.3 = 330 m s^{-1} (2 s .f.) *[1 mark]*

[Total: 3 marks]

Part (d) [3 marks]

Fig.1.1 represents the sound wave from part (c).

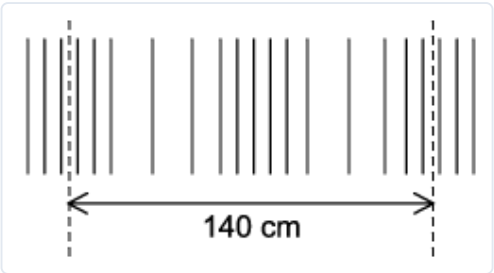


Fig. 1.1

Calculate the frequency of the sound wave.

Mark Scheme (3 marks):

To calculate the frequency of the sound wave:

List the known quantities:

- Speed of sound, v = 333 m s^{-1} (from part (c))

Determine the wavelength:

- From Fig. 1.1: 140 cm = 2λ
- λ= 70 cm *[1 mark]*

Calculate the frequency:

- $f = \frac{v}{\lambda}$
- $f = \frac{333}{0.70}$*[1 mark]*
- f = 476 = 480 Hz (2 s. f.) *[1 mark]*

[Total: 3 marks]

Question 5 Medium [9 marks]

Part (a) [2 marks]

A large water tank is set up so that a wave can be generated at each end of the tank. The two waves, A and B, travel towards each other at the same speed.

Fig. 1.1 shows the variation of displacement of the water surface with distance travelled at a particular instant.

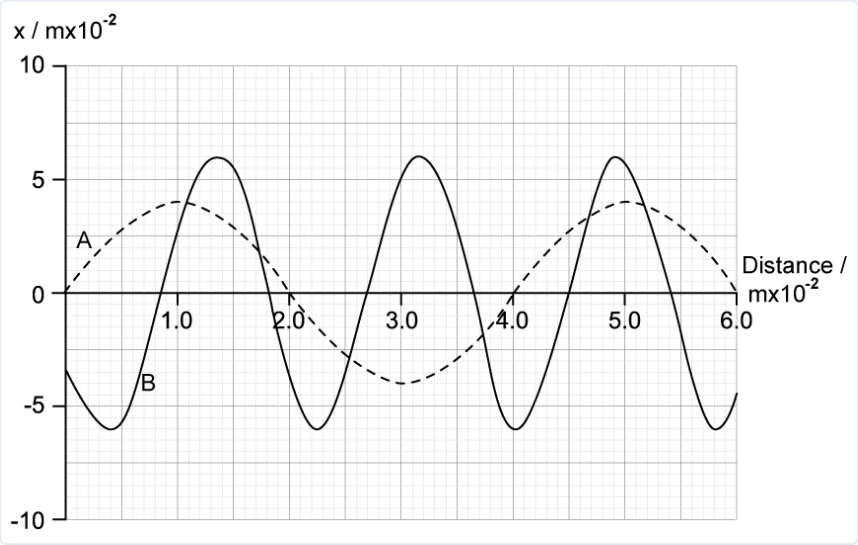


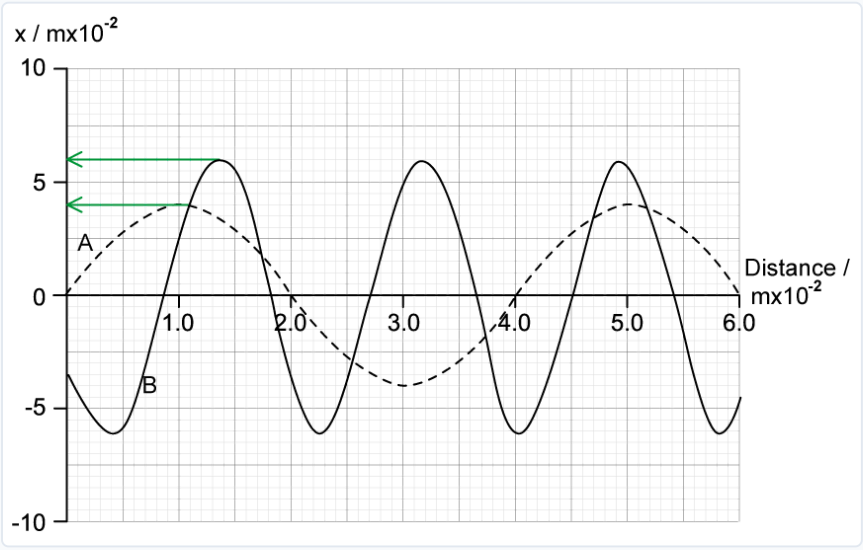
Fig. 1.1

Deduce how many times greater the amplitude of B is to the amplitude of A.

Mark Scheme (2 marks):

Determine the amplitudes from the graphs:

- In each case reading maximum displacement from the *y*-axis of the graph:



- Wave A = 4 cm **AND** Wave B = 6 cm; *[1 mark]*

Calculate how much B is larger than A:

- $\frac{B}{A} = \frac{6}{4} = \frac{3}{2}$
- B is $\frac{3}{2}$ **OR** 1.5 times larger than A; *[1 mark]*

[Total: 2 marks]

Part (b) **[4 marks]**

Wave A has a frequency of 8.0 Hz.

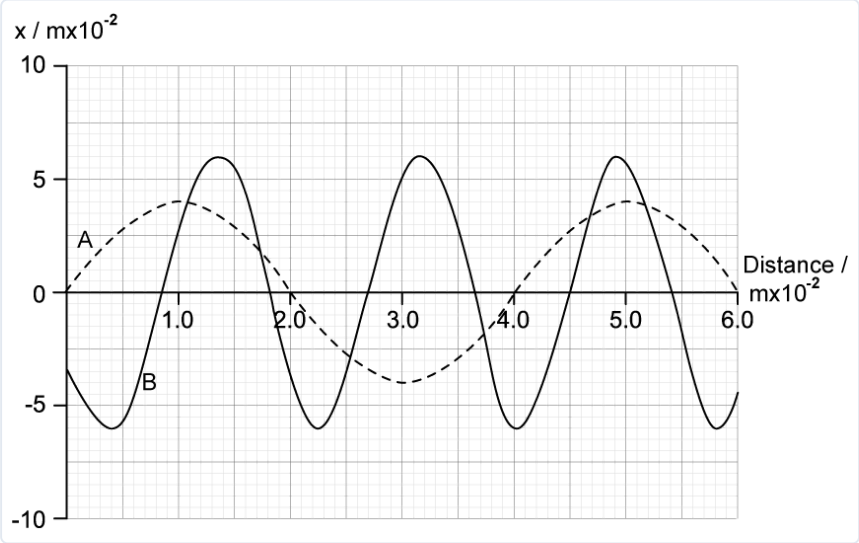


Fig. 1.1

(i) Calculate the velocity of wave A.

[2]

(ii) Determine the frequency of wave B.

[2]

Mark Scheme (4 marks):

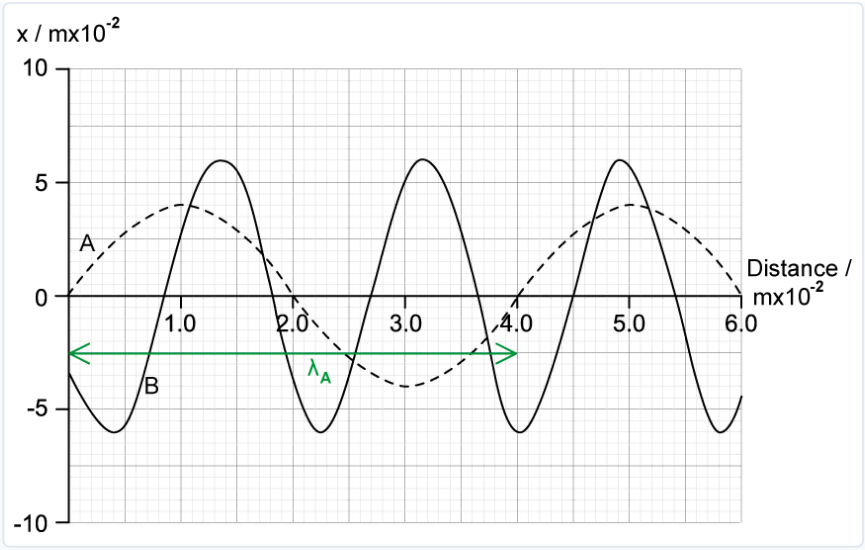
(i) The velocity of wave A:

List the known quantities:

- Frequency A, f_A = 8.0 Hz

Determine wavelength of A from the graph:

- Wavelength A, λ_A = 4.0 cm **OR** = 0.04 m *[1 mark]*
    - You can read wavelength it from the x-axis, at the point where one wave completes



Calculate the speed of the wave:

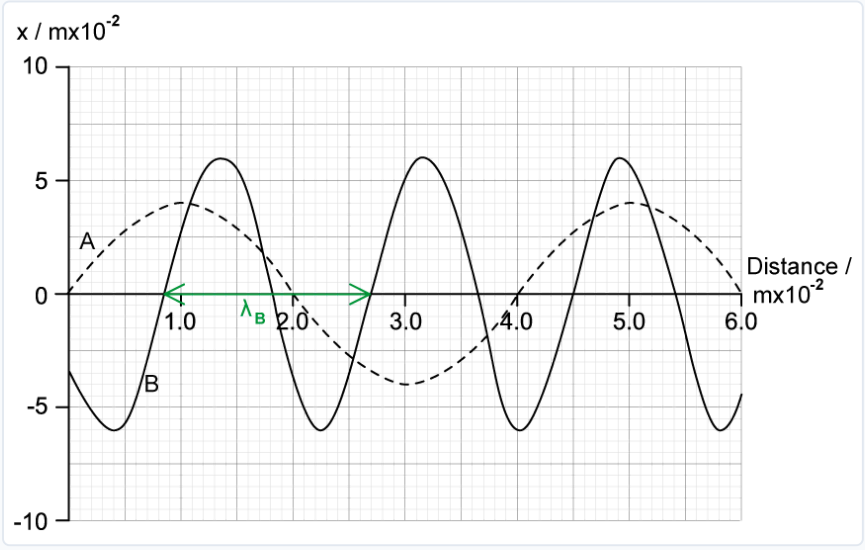
- velocity of wave A:
  - $v = f\lambda$
  - $v_A = f_A\lambda_A = 4.0 \times 8.0$
  - v = 32 cm s^{-1} **OR** = 0.32 m s^{-1} *[1 mark]*

(ii) Determine the frequency of wave B:

List the known quantities:

- Velocity of wave A, v_A= 32 cm s^{-1}

Determine wavelength of B from the graph:



- Wavelength B, λ_B = 1.8 cm **OR** = 0.018 m *[1 mark]*

Determine velocity of wave B:

- Velocity of wave B = velocity wave A, v_B = 32 cm s^{-1} (because they are travelling at the same speed)

Calculate the frequency of the wave:

- $f_B = \frac{v_B}{\lambda_B} = \frac{32}{1.8}$
- f_B = 18 Hz *[1 mark]*

[Total: 4 marks]

In a question like this, where measurements are given in non-SI units, you must state your units in your final answer. It is reasonable to calculate for these waves in either centimetres or metres, so be careful to show which you worked in and clearly show in your answer which is applicable. Notice that in the final calculation for frequency, as long as you do not mix up your units, i.e.: use all centimetres or all metres, then the value for frequency will be correct as the unit of length has cancelled out as this is a ratio.

Part (c) **[3 marks]**

Explain how the stationary wave is generated in the tank.

Mark Scheme (3 marks):

The stationary wave is generated in the tank by:

- Two (water) waves (in the tank) move in opposite directions; *[1 mark]*
- As they are reflected from the fixed ends of the tank; *[1 mark]*
- The two waves interfere / superpose to produce a standing / stationary wave; *[1 mark]*

[Total: 3 marks]

The key words here are the 'interfere' or 'superpose' for how the waves 'meet' to create the stationary wave. It is better to use these terms instead of just 'meet' or 'overlap' as they are more relevant to this scenario and not as vague. This makes it much easier for the examiner to award you the mark!